MIL-N-3337C
14 JUNE 1985
Superseding
MIL-N-3337B
2 Aug 1966

MILITARY SPECIFICATION

NUT, SHEET SPRING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification controls requirements of nuts manufactured from sheet metal with an aperture capable of receiving, gripping and securely holding threaded fasteners and related items.

1.2 Classification. Sheet spring nuts shall be furnished in one of the following types, and shall be of the screw sizes listed in Table I and Table II, as specified (see 3.1.2 and 6.2):

Type I - For use with coarse-thread series machine screws.

Type II - For use with (sheet metal) thread-forming screws.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

	FF-S-92	Screw, Machine, Slotted, Cross-Recessed or Hexagon Head
*	FF-S-107	Screw, Tapping and Drive
	QQ-P-416	Plating, Cadmium (Electrodeposited)
	TT-C-490	Cleaning Method and Pre-Treatment of Ferrous Surfaces for Organic Coatings
*	PPP-H-1581	Hardware (Fasteners and Related Items) Packaging of

MILITARY

*	DOD-D-1000	Drawing, Engineering and Associated List
*	MIL-M-6874	Metal Spraying, Process For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-66	Steel, Chemical Composition and Hardenability
FED-STD-151	Metal, Test Methods

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	MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
	MIL-STD-130	Identification Marking of U.S. Military Property
	MIL-STD-129	Marking for Shipment and Storage
*	MIL-STD-147	Palletized Unit Loads
*	MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
	MS90723	Nut, Sheet Spring "J" Type
	MS90724	Nut, Sheet Spring "U" Type

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design. The sheet spring nut shall consist of one piece, designed for engagement and locking with standard coarse-thread machine or thread-forming screws. The machine screw threads shall conform to the American National coarse-thread Class 2B fit. The thread-forming screw threads shall conform to self-tapping threads of FF-S-107. The sheet spring nut shall conform to the type, dimensions, material, finish and size of Military Standards MS90723 or MS90724, or as specified in the contract or order.

3.1.1 The form of element incorporating the sheet spring nut feature shall be in conformance with pertinent Government, industry, or vendor drawings, as specified by the procuring activity. (see 6.1).

3.1.2 Size. The nuts shall be made in various sizes for use with standard screws; Type I for coarse-thread machine screws as specified in Table I, and Type II for thread forming screws as specified in Table II (see 6.2).

3.1.3 Contact edges. All edges shall be burred or so formed that they will not damage the surface contacted in assembly.

3.2 Performance.

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TABLE I. Requirements data for type I nuts

Column							
1	2	3	4	5	6	7	8
Screw size	Torque minimum lb in.	Failure torque minimum lb in.	Tensile test load pounds	Failure load minimum pounds	Frequency cpm	Vibration double amplitude inch	Vibration load pounds
4	2	3	65	98	3,600	1/32	16
6	5	7	110	165	3,600	1/16	25
8	8	10	150	225	3,600	1/16	38
10	14	16	220	330	3,600	1/16	50
1/4	25	35	310	465	3,600	1/16	85

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TABLE II, Requirements data for Type II nuts.

Column							
1	2	3	4	5	6	7	8
Screw Size	Torque minimum lb in.	Failure torque minimum lb in	Tensile test load pounds	Failure load minimum pounds	Frequency cpm	Vibration double amplitude inch	Vibration load pounds
4	6	8	200	300	3,600	1/32	38
6	10	12	260	390	3,600	1/16	70
8	14	18	340	510	3,600	1/16	95
10	25	30	500	750	3,600	1/16	135
1/4	50	60	780	1,170	3,600	1/16	185

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3.2.1 Minimum torque. A screw assembled to the spring nut with the minimum torque specified in Table I and Table II shall not loosen when subjected to the vibration requirements of column 6, 7, and 8 of Tables I and II.

3.2.2 Failure torque. The nuts shall not fail at a torque load of less than the values given in column 3 of Tables I and II.

3.2.3 Tension. The nuts shall withstand a pull normal to the plane of the sheets of not less than the tensile load given in column 4 of Tables I and II, without permanent set. The nuts shall not fail when a pull normal to the plane of the sheets less than the tensile load given in column 5 of Tables I and II is applied.

3.2.4 Vibration. Sheet spring nuts shall withstand being vibrated with a simple harmonic motion, without any evidence of failure, for at least 20 hours at the frequency, double amplitude, and loads given in Tables I and II, when assembled with the minimum torques specified in Tables I and II.

3.2.5 Endurance. Sheet spring nuts shall withstand without any evidence of failure, 15 cycles of tightening and loosening operations as specified in 4.3.6. At the end of this test, the nut screw thread, locking element, spring devices, or other parts of the assembly shall not be distorted or damaged in any manner that might affect subsequent use.

3.3 Finish. Noncorrosion-resistant metals shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3. For applications not involving electrical bonding, noncorrosion-resistant metals may be alternately treated with a heavy phosphate coating conforming to TT-C-490, Grade A, followed by a prime coat of zinc-chromate primer conforming to TT-P-1757.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.5 Identification of product. All nuts, where practicable, shall be marked as to the thread size, type of screw used, and manufacturer's identification thereof in accordance with the requirements of MIL-STD-130.

3.6 Workmanship. All details of workmanship shall be in accordance with highgrade manufacturing practice. The nuts shall be free from tool marks, burrs, flaws, and all other defects.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105, Inspection Level II and AQL of 6.5 percent defective. Sampling for the Torque Tension, Vibration, Corrosion, and Endurance tests specified in accordance with Inspection Level L4 of the appendix to MIL-STD-105 and AQL of 2.5 percent defective.

4.2.1 All screws and fixtures necessary for the tests specified herein shall be furnished by the contractor at no cost to the Government. Screws shall be in accordance with 3.1.

4.3 Inspection and tests.

4.3.1 Inspection. The sheet spring nuts shall be examined for conformance with this specification with respect to material, workmanship, configuration, contact edges, marking, and finish.

4.3.2 Torque. All sample nuts subjected to these tests shall be mounted on a test fixture as shown on Figure 1. The screw shall be tightened in the nut to the values shown in column 2 of Tables I and II. Steel screws shall be used in testing steel sheet spring nuts and shall have dry threads. Corrosion-resistant steel screws shall be used in testing corrosion-resistant steel sheet spring nuts and shall have dry threads.

4.3.3 Tension. The tension test shall be conducted on a test fixture as shown on Figure 2. The screw shall be tightened in the nut to the installation torque shown in columns 2 of Table I and II, as applicable, and the test jig assembled in a suitable test machine and subjected to the Tensile test load and failure load shown in columns 4 and 5.

4.3.4 Vibration. Sheet spring nuts shall be mounted as shown on Figure 3, and subjected to the vibration requirements of Tables I and II for the specified period of time. The screws shall be assembled with the minimum torques specified in Tables I and II.

4.3.5 Corrosion. The sheet spring nuts shall withstand salt spray tests in accordance with FED-STD-151 for 192 hours without indications of corrosion which cannot be wiped off with a damp cloth. For purposes of this requirement, a corroded area is defined as exposure or corrosion of the basic metal.

4.3.6 Endurance. The nut shall be tested on the test fixture shown on Figure 1. The screw shall be tightened in the nut to the values shown in column 2 of Tables I and II. The screw shall then be loosened. These operations constitute one complete cycle.

4.4 Uniformity. Sheet spring nuts supplied under contract shall be identical in every respect to the sample tested and found satisfactory, and to the detail drawings thereof, except for changes previously approved by the Government. Any unapproved changes from the accepted test sample or from the drawings shall constitute cause for rejection.

* 4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of Paragraph 5.1, 5.2, and 5.3 of this specification.

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5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A, C or Commercial as specified (6.2.1).

* 5.1.1 Level A. The sheet spring nuts shall be preserved in accordance with PPP-H-1581.

* 5.1.2 Level C. The Level C preservation for the sheet spring nuts shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of the sheet spring nuts shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.2.1).

* 5.2.1 Level A and B. The sheet spring nuts shall be packed for shipment in accordance with PPP-H-1581, except fiberboard boxes shall not be used for Level A packing.

* 5.2.2 Level C. The level C packing for the sheet spring nuts shall conform to the MIL-STD-794 requirements this level.

* 5.2.3 Commercial. The preserved sheet spring nuts shall be packed in accordance with the requirements of ASTM D 3951.

* 5.2.5 Palletization. When specified (See 6.2.1) unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Unitized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.3 Marking.

* 5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract (See 6.2.1) each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture or date vulcanized for the type of hose specified in the contract or order.

* 5.3.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM D 3951.

* 5.4 General.

* 5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.3, and 5.2.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

* 5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

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6. NOTES

6.1 Intended Use. The sheet spring nuts controlled by this specification are for other than aircraft primary or secondary structural applications and for use in fastening together of nonstructural items.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the follows:

a. Title, number and date of this specification.
b. Type and size required. (See 3.1.2)
c. Material. (See 3.1)
d. Design. (See 3.1)
e. Finish required (See 3.3)
f. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial (See 5.1, 5.2 and 5.3)
g. When palletized loads are required (see 5.4)

6.3 For Bureau of Ships and Air Force use, the limitations of Standard MS33538 should be considered.

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issued.

Custodians:

Army - AV
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Air Force - 99

Review Activity:

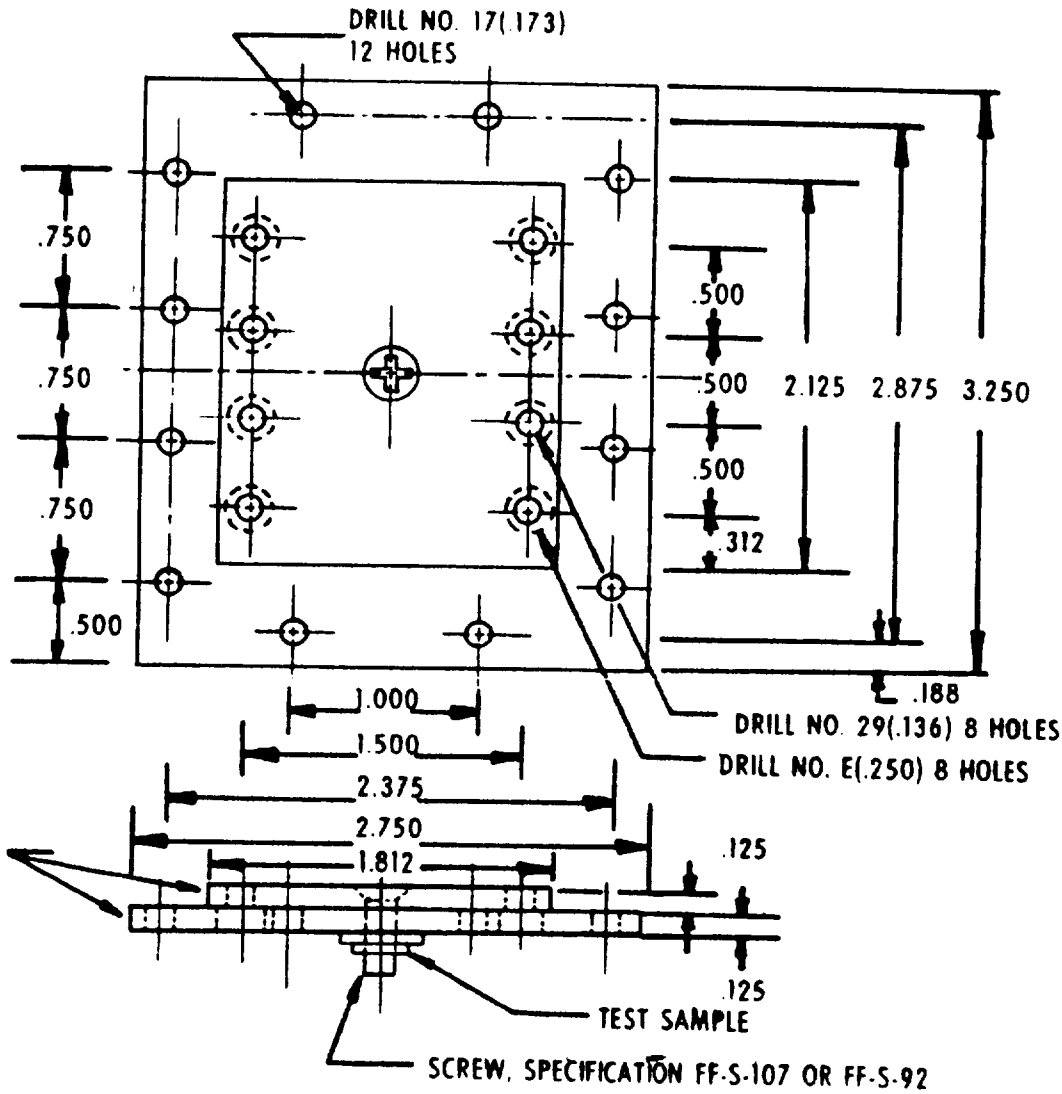
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Preparing Activity:

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DIMENSIONS IN INCHES. TOLERANCE: DECIMAL +- .016

FIGURE 1. TEST FIXTURE FOR MOUNTING SPRING NUTS

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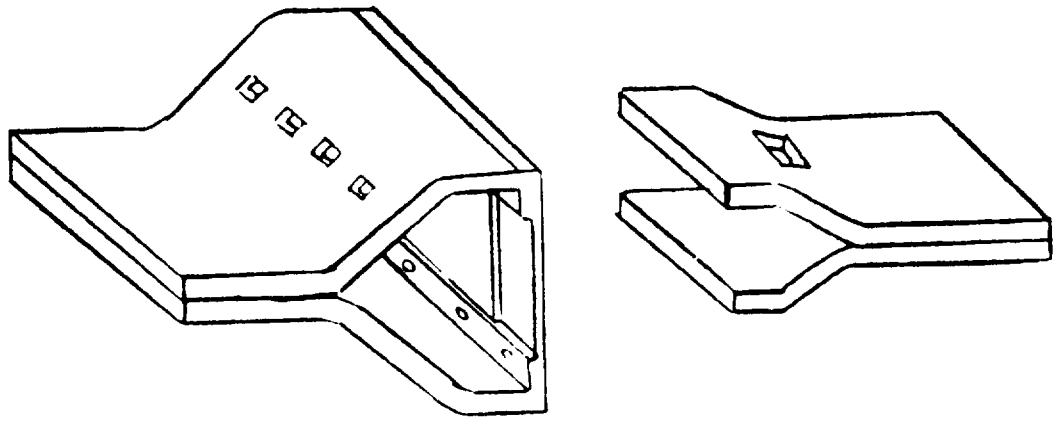


FIGURE 2. TEST JIG

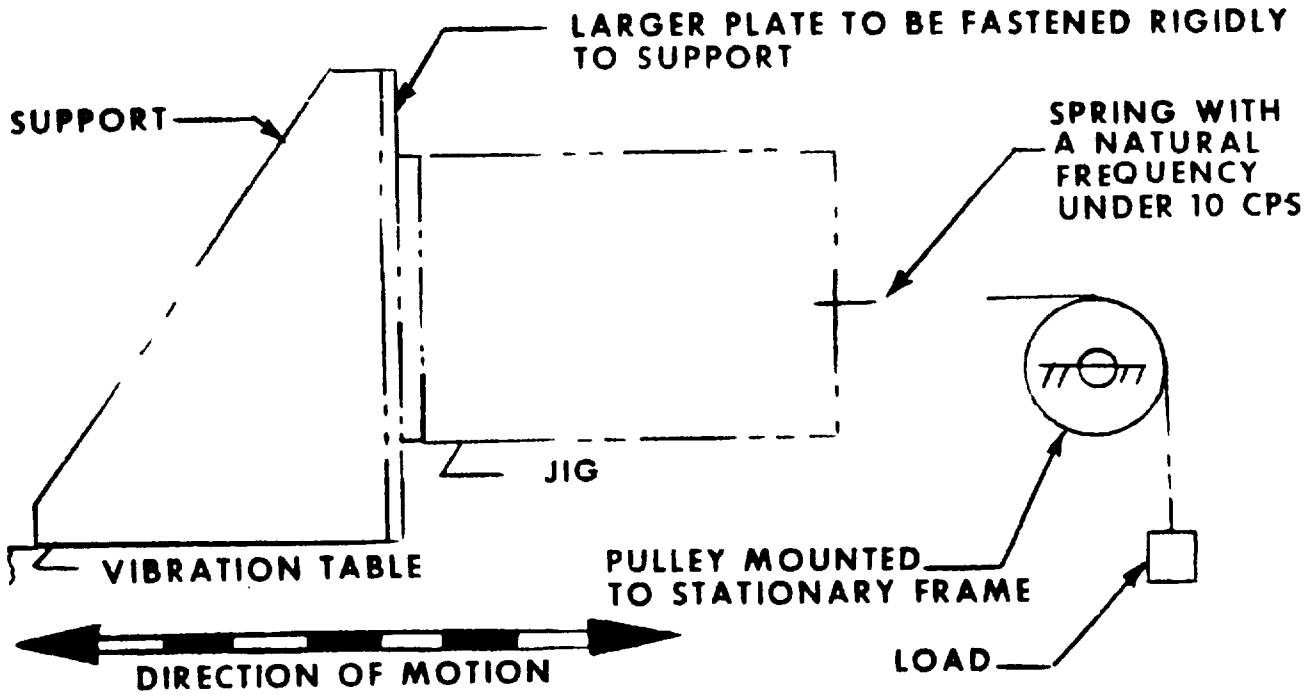


FIGURE 3. VIBRATION TEST GROUP

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